

Quantum Mechanics I Exercise 2

Idan Stark

November 24, 2025

1 The relation between the propagator perturbation expansion in path integral and operator level

1.1 Schrodinger equation in the interaction representation

Considering the Schrodinger equation defined by

$$i\hbar \frac{\partial}{\partial t} \psi(t) = (\hat{H}_0 + \hat{V}) \psi(t) \quad (1)$$

Let us define the interaction wave function as

$$\varphi(t) = e^{i\hat{H}_0 t/\hbar} \psi(t) \quad (2)$$

Where $\psi(t)$ is the wave function solving equation (1). Solving equation (2) for $\psi(t)$ and assigning it into equation (1) gives

$$\begin{aligned} i\hbar \frac{\partial}{\partial t} e^{-i\hat{H}_0 t/\hbar} \varphi(t) &= (\hat{H}_0 + \hat{V}) e^{-i\hat{H}_0 t/\hbar} \varphi(t) \\ \hat{H}_0 e^{-i\hat{H}_0 t/\hbar} \varphi(t) + i\hbar e^{-i\hat{H}_0 t/\hbar} \frac{\partial \varphi(t)}{\partial t} &= (\hat{H}_0 + \hat{V}) e^{-i\hat{H}_0 t/\hbar} \varphi(t) \\ i\hbar e^{-i\hat{H}_0 t/\hbar} \frac{\partial \varphi(t)}{\partial t} &= \hat{V} e^{-i\hat{H}_0 t/\hbar} \varphi(t) \\ i\hbar \frac{\partial \varphi(t)}{\partial t} &= e^{i\hat{H}_0 t/\hbar} \hat{V} e^{-i\hat{H}_0 t/\hbar} \varphi(t) \end{aligned} \quad (3)$$

And so we get the required equation where

$$\hat{V}_I(t) = e^{i\hat{H}_0 t/\hbar} \hat{V} e^{-i\hat{H}_0 t/\hbar} \quad (4)$$

Other operators would be represented in a similar manner:

$$\hat{\Omega}_I(t) = e^{i\hat{H}_0 t/\hbar} \hat{\Omega} e^{-i\hat{H}_0 t/\hbar} \quad (5)$$

1.2 Time evolution operator in the interaction representation

The time evolution operator satisfies the equation

$$|\psi(t)\rangle = \hat{U}_I(t, t_0) |\psi(t_0)\rangle \quad (6)$$

Putting this into the interaction representation Schrodinger equation, we get

$$\begin{aligned} i\hbar \frac{\partial}{\partial t} \hat{U}_I(t, t_0) |\psi(t_0)\rangle &= \hat{V}_I(t) \hat{U}_I(t, t_0) |\psi(t_0)\rangle \\ i\hbar \frac{\partial}{\partial t} \hat{U}_I(t, t_0) &= \hat{V}_I(t) \hat{U}_I(t, t_0) \end{aligned} \quad (7)$$

This is a simple ODE, the solution to which can be achieved using the Dyson Series

$$\hat{U}_I(t, t_0) = \mathcal{T} \exp \left[-\frac{i}{\hbar} \int_{t_0}^t \hat{V}_I(\tau) d\tau \right] \quad (8)$$

Since equation (5) is true for any operator, it's true in particular to $U(t_f, t_i)$. We multiply by the exponent from both sides to get:

$$\hat{U}(t_f, t_i) = e^{-i\hat{H}_0 t_f / \hbar} \hat{U}_I(t_f, t_i) e^{i\hat{H}_0 t_i / \hbar} \quad (9)$$

1.3 The Perturbation expansion of $\hat{U}(t_f, t_i)$

Openning equation (8) back into its exponent expansion, we get:

$$\begin{aligned} \hat{U}(t_f, t_i) &= e^{-i\hat{H}_0 t_f / \hbar} \mathcal{T} \exp \left[-\frac{i}{\hbar} \int_{t_0}^t \hat{V}_I(\tau) d\tau \right] e^{i\hat{H}_0 t_i / \hbar} = \\ &= \sum_{n=0}^{\infty} \frac{(-i)^n}{\hbar^n n!} \int \prod_{k=1}^n dt_k \mathcal{T} \left[\prod_{k=1}^n e^{-i\hat{H}_0(t_f - t_k) / \hbar} \hat{V}(t_k) e^{-i\hat{H}_0(t_k - t_i) / \hbar} \right] \end{aligned} \quad (10)$$

Where $t_n = t_f$ and $t_0 = t_i$. In other words, we can write $\hat{U}_I(t_f, t_i)$ as

$$\hat{U}_I(t_f, t_i) = 1 + \frac{i}{\hbar} \int_{t_i}^{t_f} \hat{V}_I(\tau) d\tau + \dots \quad (11)$$

And so $\hat{U}(t_f, t_i)$ is

$$\hat{U}(t_f, t_i) = e^{-i\hat{H}_0 t_f / \hbar} \left(1 + \frac{i}{\hbar} \int_{t_i}^{t_f} \hat{V}_I(\tau) d\tau + \dots \right) e^{i\hat{H}_0 t_i / \hbar} \quad (12)$$

The zeroth order term is

$$U_0(t_f, t_i) = e^{-i\hat{H}_0(t_f - t_i) / \hbar} \quad (13)$$

So we can rewrite equation (10) as

$$\hat{U}(t_f, t_i) = \sum_{n=0}^{\infty} \frac{(-i)^n}{\hbar^n n!} \int \prod_{k=1}^n dt_k \mathcal{T} \left[\prod_{k=1}^n \hat{U}_0(t_f, t_k) \hat{V}(t_k) \hat{U}_0(t_k, t_i) \right] \quad (14)$$

The first and second order terms are

$$U_1(t_f, t_i) = -\frac{i}{\hbar} \int dt_1 \hat{U}_0(t_f, t_1) \hat{V}(t_1) \hat{U}_0(t_1, t_i) \quad (15)$$

$$\begin{aligned} U_2(t_f, t_i) &= -\frac{1}{\hbar^2} \int dt_1 dt_2 \hat{U}_0(t_f, t_2) \hat{V}(t_2) \hat{U}_0(t_2, t_1) \hat{V}(t_1) \hat{U}_0(t_1, t_i) = \\ &= -\frac{i}{\hbar} \int dt_2 \hat{U}_0(t_f, t_2) \hat{V}(t_2) \hat{U}_1(t_2, t_i) \end{aligned} \quad (16)$$

Which is a similar relation to the one we got in equation (90) in the course notes:

$$U_k(t_f, t_i) = -\frac{i}{\hbar} \int d\tau \hat{U}_0(t_f, \tau) \hat{V}(\tau) \hat{U}_{k-1}(\tau, t_i) \quad (17)$$

1.4 The full integral equation for $\hat{U}(t_f, t_i)$

We can use the recursion relation that can be seen in equation (17) to find the full $\hat{U}(t_f, t_i)$ by simply assigning it into equation (14), remembering that everything inside the sum is U_n :

$$\begin{aligned}
\hat{U}(t_f, t_i) &= \sum_{n=0}^{\infty} U_n(t_f, t_i) = \\
&= U_0 + \sum_{n=1}^{\infty} U_n(t_f, t_i) = \\
&= U_0 + \sum_{n=0}^{\infty} U_{n+1}(t_f, t_i) = \\
&= U_0 + \sum_{n=0}^{\infty} -\frac{i}{\hbar} \int d\tau \hat{U}_0(t_f, \tau) \hat{V}(\tau) \hat{U}_n(\tau, t_i) = \\
&= U_0 - \frac{i}{\hbar} \int d\tau \hat{U}_0(t_f, \tau) \hat{V}(\tau) \sum_{n=0}^{\infty} \hat{U}_n(\tau, t_i) = \\
&= U_0 - \frac{i}{\hbar} \int d\tau \hat{U}_0(t_f, \tau) \hat{V}(\tau) \hat{U}(\tau, t_i)
\end{aligned} \tag{18}$$

2 The propagator in momentum space

Given a Hamiltonian $H_{op}(p_{op}, x_{op})$, the propagator

$$U(t_f, t_i) = e^{-iH_{op}(t_f - t_i)/\hbar} \quad (19)$$

can be represented in the different bases by slicing it. Let

$$\epsilon = \frac{t_f - t_i}{N} \quad (20)$$

And let $\{|q_n\rangle\}$ be a complete bases

$$\int |q_n\rangle \langle q_n| dq_n = 1 \quad (21)$$

We can represent $U(t_f, t_i)$ in the $\{|q_n\rangle\}$, and insert the unit from equation (12) into the definition of U , slicing it into N smaller propagators. Using the momentum base, this becomes:

$$\begin{aligned} K(p_f, t_f; p_i, t_i) &= \langle p_f | U(t_f, t_i) | p_i \rangle = \\ &= \int dp_1 dp_2 \dots dp_{N-1} \langle p_f | e^{-iH_{op}\epsilon/\hbar} | p_{N-1} \rangle \langle p_{N-1} | e^{-iH_{op}\epsilon/\hbar} | p_{N-2} \rangle \dots \langle p_2 | e^{-iH_{op}\epsilon/\hbar} | p_i \rangle \end{aligned} \quad (22)$$

Where $|p_f\rangle = |p_N\rangle$ and $|p_i\rangle = |p_1\rangle$.

For the simple Hamiltonian

$$H_{op} = \frac{p_{op}^2}{2m} + V(x_{op}) \quad (23)$$

we can expand each factor in the integral using exponent Taylor expansion. At the limit where $N \rightarrow \infty, \epsilon \rightarrow 0$, the commutator of the kinetic and potential terms of the Hamiltonian gets a $O(\epsilon^2)$ in the exponent expansion, and since there are N such factor get a total $O(\epsilon)$ which is negligible. We can then write the factors as

$$e^{-iH_{op}\epsilon/\hbar} = e^{-\frac{i\epsilon}{\hbar} V(x_{op})} e^{-\frac{i\epsilon}{\hbar} \frac{p_{op}^2}{2m}} \quad (24)$$

Applying this to the p_{n-1} state and inserting the $|x_n\rangle$ completion identity to evaluate $V(x_{op})$, we get:

$$\begin{aligned} \langle p_n | e^{-i\epsilon H_{op}/\hbar} | p_{n-1} \rangle &= \\ &= e^{-\frac{i\epsilon}{\hbar} \frac{p_n^2}{2m}} \langle p_n | e^{-\frac{i\epsilon}{\hbar} V(x_{op})} | p_{n-1} \rangle = \\ &= \frac{1}{2\pi\hbar} e^{-\frac{i\epsilon}{\hbar} \frac{p_n^2}{2m}} \int dx_n \langle p_n | e^{-\frac{i\epsilon}{\hbar} V(x_{op})} | x_n \rangle \langle x_n | p_{n-1} \rangle = \\ &= \frac{1}{2\pi\hbar} e^{-\frac{i\epsilon}{\hbar} \frac{p_n^2}{2m}} \int e^{-\frac{i\epsilon}{\hbar} V(x_n) - i(p_n - p_{n-1})x_n/\hbar} dx_n = \\ &= \frac{1}{2\pi\hbar} \int e^{-\frac{i\epsilon}{\hbar} H(p_n, x_n) + i(p_n - p_{n-1})x_n/\hbar} dx_n \end{aligned} \quad (25)$$

This can be expanded to a generic Hamiltonian. The steps are similar, with the one change being that the expansion of the exponent is more direct:

$$\begin{aligned} \langle p_n | e^{-i\epsilon H_{op}(p_{op}, x_{op})/\hbar} | p_{n-1} \rangle &= \\ &= \frac{1}{2\pi\hbar} \int dx_n \langle p_n | e^{-i\epsilon H_{op}(p_{op}, x_{op})/\hbar} | x_n \rangle \langle x_n | p_{n-1} \rangle = \\ &= \frac{1}{2\pi\hbar} \int dx_n \langle p_n | e^{-i\epsilon H_{op}(p_n, x_n)/\hbar} | x_n \rangle \langle x_n | p_{n-1} \rangle = \\ &= \frac{1}{2\pi\hbar} \int e^{-\frac{i\epsilon}{\hbar} H(p_n, x_n) + i(p_n - p_{n-1})x_n/\hbar} dx_n \end{aligned} \quad (26)$$

The result, in both cases, when accumulating all factors in the equation (13) integral, is:

$$K(p_f, t_f, p_i, t_i) = \int \prod_{n=2}^N \frac{dx_n}{2\pi\hbar} \prod_{n=2}^N dp_n \exp \left[\frac{i\epsilon}{\hbar} \sum_{n=2}^N \left(x_n \frac{p_n - p_{n-1}}{\epsilon} - H(p_n, x_n) \right) \right] \quad (27)$$

Where the factor $(2\pi\hbar)$ is the differential volume element. This is very similar to the position representation of the propagator. The key difference is in the first term of the sum in the exponent. If in the position representation propagator, this term reminded us of a Legendre transform according to position $p\dot{x} - H$, this time it reminds us of a Legendre transform according to the momentum $x\dot{p} - H$.

When the Hamiltonian is dependent only on the momentum $H_{op} = H(p_{op})$, we get:

$$\begin{aligned} K(p_f, t_f, p_i, t_i) &= \int \prod_{n=2}^N \frac{dx_n}{2\pi\hbar} \prod_{n=2}^N dp_n \exp \left[\frac{i\epsilon}{\hbar} \sum_{n=2}^N \left(x_n \frac{p_n - p_{n-1}}{\epsilon} - H(p_n) \right) \right] = \\ &= \left(\frac{1}{2\pi\hbar} \right)^N \int \prod_{n=2}^N dx_n \prod_{n=2}^N dp_n \exp \left[\frac{i}{\hbar} \sum_{n=2}^N x_n (p_n - p_{n-1}) \right] \exp \left[-\frac{i\epsilon}{\hbar} \sum_{n=2}^N H(p_n) \right] \end{aligned} \quad (28)$$

For every integral position integral, we get:

$$\int dx_n \exp \left[\frac{i}{\hbar} x_n (p_n - p_{n-1}) \right] = \delta(p_n - p_{n-1}) \quad (29)$$

Therefore:

$$\begin{aligned} K(p_f, t_f, p_i, t_i) &= \left(\frac{1}{2\pi\hbar} \right)^N \int \prod_{n=2}^N \left[\delta(p_n - p_{n-1}) e^{-\frac{i\epsilon}{\hbar} H(p_n)} dp_n \right] = \\ &= \left(\frac{1}{2\pi\hbar} \right)^N \int \delta(p_2 - p_1) \delta(p_3 - p_2) \cdots \delta(p_N - p_{N-1}) e^{-\frac{i\epsilon}{\hbar} H(p_2)} e^{-\frac{i\epsilon}{\hbar} H(p_3)} \cdots e^{-\frac{i\epsilon}{\hbar} H(p_f)} dp_2 dp_3 \cdots dp_N = \\ &= \left(\frac{1}{2\pi\hbar} \right)^N e^{-\frac{i\epsilon}{\hbar} N H(p_i)} \end{aligned} \quad (30)$$

3 Morse index

We can think of the morse index as the number of directions in which the second variation operator is a "sad" parabola (i.e. ≤ 0), and thus a local maximum instead of minimum. If every eigenvalue in the second variation operator are positive, this means it's a minimum in all directions, and therefore a minimum. If some are positive and some are negative, it is a saddle point, and if all are negative, it means it's a maximum.

Therefore, for classical paths, each one is either a minimum or a saddle point, depending on whether the second variation operator has negative eigenvalues, meaning if there exists focal points on the path.

As the path becomes a saddle point near the focal point, it makes sense that it would interact with another path. This is what happens when x_f approaches the caustic. The two paths approaching x_f get closer and closer until, when x_f is on the caustic, one eigenvalue from each path vanishes as the paths merge locally to form a saddle point.

When x_f is in a classically forbidden area, we must cross a focal point, adding a phase to the trajectory. As indicated, this makes the trajectory into a saddle point, and not a minimum, meaning that while the action is stationary it is not a stable solution to the Jacobi equation.